EXPT NO: 8	KUHN-TUCKER CONDITIONS LAGRANGIAN MULTIPLIER METHOD
DATE:	

AIM:

The objective of this lab is to introduce and implement the Kuhn-Tucker conditions for constrained optimization using Python. Participants will learn the theoretical basis of the Kuhn-Tucker conditions and apply them to solve optimization problems with inequality constraints.

PROCEDURE:

a. Objective Function and Constraints:

● Define the objective function and inequality constraints in Python.

b. Kuhn-Tucker Conditions:

● Implement the Kuhn-Tucker conditions using Python functions.

● Compute the gradients of the objective function and constraints.

c. Optimization Setup:

● Use scipy.optimize.minimize or another suitable optimization function.

● Incorporate the Kuhn-Tucker conditions as additional constraints or through custom callback functions.

d. Running the Optimization:

● Execute the optimization procedure and monitor convergence.

PROGRAM:

```
import numpy as np
from scipy.optimize import minimize

def objective(x):
    return x[0]**2 + x[1]**2

# x + y >= 1  →  x + y - 1 >= 0
def constraint1(x):
    return x[0] + x[1] - 1

# x >= 0
def constraint2(x):
    return x[0]

# y >= 0
def constraint3(x):
    return x[1]

# Gradient of objective
def grad_f(x):
    return np.array([2*x[0], 2*x[1]])

# Gradient of constraint (x + y - 1)
def grad_g(x):
    return np.array([1, 1])

constraints = [
    {'type': 'ineq', 'fun': constraint1},
    {'type': 'ineq', 'fun': constraint2},
    {'type': 'ineq', 'fun': constraint3}
]

# Initial guess
x0 = [0.5, 0.5]
```

```
# Solve
result = minimize(objective, x0, constraints=constraints)

print("Optimal solution:", result.x)
print("Minimum value:", result.fun)

x_opt = result.x

print("\nGradient of objective at optimum:", grad_f(x_opt))
print("Constraint value (x + y - 1):", constraint1(x_opt))
```

OUTPUT:

```
Optimal solution: [0.5 0.5]
Minimum value: 0.5

Gradient of objective at optimum: [1. 1.]
Constraint value (x + y - 1): 0.0
```

RESULT:

Thus, **kuhn-tucker conditions lagrangian multiplier method** has been completed successfully.